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SARDAR PATEL UNIVERSITY
B.C.A (4th SEM) - EXAMINATION APRIL 2026
US04CBA52 – DATA STRUCTURES-II

**Date: 04-04-2026(Saturday)****Time: 02:00 PM to 05:00 PM****NOTE: Right hand figure indicates the marks of each question.****Total Marks: [70]****Q. 1 Multiple Choice Questions (One marks each)****[10]**

- 1 A binary tree has at most _____ child.
(A) Two (B) One
(C) Three (D) None of these
- 2 _____ is a rank of hierarchy of a tree.
(A) Degree (B) Level
(C) Height (D) Index
- 3 _____ means a link between parent and its child.
(A) Degree (B) Height
(C) Leaf (D) Branch
- 4 A linked list is a _____ type of data structure.
(A) Linear (B) Non-Linear
(C) Both (A) and (B) (D) None of above
- 5 Double linked list is also known as _____.
(A) One-way list (B) Two-way list
(C) Three-way list (D) Four-way list
- 6 Each node in singly linked list has _____ fields.
(A) 2 (B) 3
(C) 14 (D) 4
- 7 _____ is the process to arrange the data in ascending or descending order.
(A) Sorting (B) Inserting
(C) Searching (D) Updating
- 8 Which design technique is used for Quick Sort?
(A) Divide and Conqueror (B) Dynamic Programming
(C) Greedy (D) Backtracking
- 9 _____ file is also known as Random file.
(A) Serial (B) Direct
(C) Sequential (D) None of these
- 10 The lowest level of index is _____.
(A) Track index (B) Prime index
(C) Index area (D) Master index

Q. 2 Short Questions. (Attempt any ten)

- 1 Define Binary Tree. List different types of binary tree.
- 2 Explain Traversals. List the types of binary tree traversal.
- 3 Define Following Terms: Tree, Root.
- 4 List delete operations that can be performed on singly linked list.
- 5 Explain doubly linked list in brief.
- 6 Explain circular linked list in brief.
- 7 Define sorting. List out the names of sorting techniques.
- 8 Explain: Internal Sort and External Sort.
- 9 Define searching. List out the names of searching techniques.
- 10 List popular hashing functions.
- 11 Define Bucket Capacity.
- 12 List types of Chaining.

Q. 3 A] List all traversal and discuss any one traversal with algorithm. [05]

B] Explain the insertion of a node in a binary with an example. [05]

OR

Q. 3 A] Explain the representation of Binary tree. [05]

B] Explain deletion of a node from binary tree (all 3 conditions) with example. [05]

Q. 4 A] Define Linked List. List and explain types of linked lists with example. [05]

B] Write an algorithm to insert node at front position in singly linked list. [05]

OR

Q. 4 A] Write an algorithm to delete node at beginning in singly linked list. [05]

B] Write an algorithm to insert node at end position in singly linked list. [05]

Q. 5 A] List sorting techniques and explain ANY ONE of them with algorithm. [10]

OR

Q. 5 A] List searching techniques and explain ANY ONE of them with algorithm. [10]

Q. 6 A] List and explain popular hashing function with example. [05]

B] Write a detail note on structure and processing of Direct files. [05]

OR

Q. 6 A] Write a detail note on: Collision resolution techniques. [05]

B] Write a detail note on structure and processing of index sequential file supported by IBM. [05]

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